Problem 1

Employee
- name: String - idNumber : int - department: String - position: String
+ Employee(): + setName(n: String): void + setIdNumber(id: int): void + setDepartment(d: String): void + setPosition(p: String): void + getName(): String + getIdNumber(): int + getDepartment(): String + getPosition(): String

```java
public class Problem1 {
    public static void main(String[] args) {
        Employee employee1 = new Employee();
        employee1.setName("Susan Meyers");
        employee1.setIdNumber(47899);
        employee1.setDepartment("Accounting");
        employee1.setPosition("Vice President");

        Employee employee2 = new Employee();
        employee2.setName("Mark Jones");
        employee2.setIdNumber(39119);
        employee2.setDepartment("IT");
        employee2.setPosition("Programmer");

        Employee employee3 = new Employee();
        employee3.setName("Joy Rogers");
        employee3.setIdNumber(81774);
        employee3.setDepartment("Manufacturing");
        employee3.setPosition("Engineer");


        System.out.println(employee1.toString());
        System.out.println(employee2.toString());
        System.out.println(employee3.toString());
    }
}
```

```java
public class Employee {
    String name;
    int idNumber;
    String department;
    String position;
    public String getName() {...}
    public void setName(String name) {...}
    public int getIdNumber() {...}
    public void setIdNumber(int idNumber) {...}
    public String getDepartment() {...}
    public void setDepartment(String department) {...}
    public String getPosition() {...}
    public void setPosition(String position) {...}
    @Override
    public String toString() {
        return "name: " + name + ", idNumber: " + idNumber + ", department: " + department + ", position: "
                + position;
    }

}
```

Problem 2

Car
- yearModel: int - make: String - speed: int
+ Car(): + setYearModel(y: int): void + setMake(m: String): void + setSpeed(s: int): void + getYearModel(): int + getMake(): String + getSpeed(): int + accelerate(): int + brake(): int

```java
public class Problem2 {
    public static void main(String[] args) {
        Car car = new Car();
        car.setYearModel(2000);
        car.setMake("BMW");
        car.setSpeed(120);

        for(int i =0; i<5; i++){
            car.accelerator();
            System.out.print(car.getSpeed()+" ");
        }
        System.out.println("");

        for(int i =0; i<5; i++){
            car.brake();
            System.out.print(car.getSpeed()+" ");
        }

    }

}
```

```java
public class Car {
    int yearModel;
    String make;
    int speed;
    public int getYearModel() {...}
    public void setYearModel(int yearModel) {...}
    public String getMake() {...}
    public void setMake(String make) {...}
    public int getSpeed() {...}
    public void setSpeed(int speed) {...}
    public void accelerator(){
        this.speed+=5;
    }
    public void brake(){
        this.speed-=5;
    }


}
```

Problem 3

Temperature
- ftemp: double
+ Temperature() + setFtemp(f: double): void + getFtemp(): double + getCelsius(): double + getKelvin(): double

```java
public class Problem3 {
    public static void main(String[] args) {
        Temperature temperature = new Temperature();
        temperature.setFtemp(12.5);
        System.out.println("Celsius: "+temperature.getCelsius()+", Kelvin: "+temperature.getKelvin());
    }
}
```

```java
public class Temperature {
    double ftemp;

    public double getFtemp() {...}

    public void setFtemp(double ftemp) {...}
    public double getCelsius() {
        double c = (ftemp-32)/1.8;
        return c;
    }
    public double getKelvin() {
        double k = (ftemp-32)/1.8 + 273;
        return k;
    }

}
```

Problem 4

SavingAccount
- annualInterest: double - startBalance: double - totalDeposits: double -totalWithdrawls: double - totalInterest: double - bank: double
+ SavingAccount() + setAnnualInterest(a: double): void + setStartBalance(sb: double): void + setTotalDeposits(td: double): void + setTotalWithdrawls(tw: double): void + setTotalInterest(ti: double): void + setBank(bank: double): void + getAnnualInterest(): double + getStartBalance(): double + getTotalDeposits(): double + getTotalWithdrawls(): double + getTotalInterest(): double + getBank(): double + withdrawal(wd: double): void + deposit(de: double): void + calculateInterest(): void

```java
import java.util.Random;
import java.util.Scanner;

public class Problem4 {
    public static void main(String[] args) {
        Random rd = new Random();
        Scanner sc = new Scanner(System.in);

        System.out.println("annual interest rate: ");
        double annualInterest = sc.nextDouble();
        System.out.println("starting balance: ");
        double startBalance = sc.nextDouble();
        System.out.println("number of months that have passed: ");
        int numberMonths = sc.nextInt();
        SavingsAccount savingsAccount = new SavingsAccount(annualInterest, startBalance);
        for(int i=0; i<numberMonths; i++){
            System.out.println("amount deposited month "+ (i+1) +": ");
            savingsAccount.deposit(sc.nextDouble());
            System.out.println("amount withdrawn: ");
            savingsAccount.withdrawal(sc.nextDouble());
            savingsAccount.calculateInterest();
        }

        System.out.println();
        System.out.println(savingsAccount.toString());
    }
}
```

```java
public class SavingsAccount {
    double annualInterest;
    double startBalance;

    double totalDeposits =0;
    double totalWithdrawals =0;
    double totalInterest= 0;
    double bank;
    public SavingsAccount(double annualInterest, double startBalance) {
        this.annualInterest = annualInterest;
        this.startBalance = startBalance;
        this.bank=startBalance;
    }

    public void withdrawal(double money){
        this.totalWithdrawals+=money;
        this.bank-=money;
    }
    public void deposit(double money){
        this.totalDeposits+=money;
        this.bank+=money;
    }
    public void calculateInterest(){
        double interest = bank*annualInterest/12;
        this.bank+=interest;
        this.totalInterest+=interest;
```

```java
    this.bank+=interest;
    this.totalInterest+=interest;
}

public double getTotalDeposits() {
    return totalDeposits;
}

public double getTotalWithdrawals() {
    return totalWithdrawals;
}

public double getTotalInterest() {
    return totalInterest;
}

public double getBank() {
    return bank;
}

@Override
public String toString() {
    return "ending balance: " + bank + "\ntotal amount of deposits: " + totalDeposits + "\ntotal amount of withdrawals: " + t
            + "\ntotal interest earned: " + totalInterest ;
}
```

Problem 5

```java
public class Problem5 {
    public static void main(String[] args) {
        Customer5 customer = new Customer5();
        customer.setName("a");
        customer.setAddress("123/1 binh duong");
        customer.setPhoneNumber("0912345678");
        Pet pet = new Pet();
        pet.setBreek(10);
        pet.setAge(5);
        pet.setWeight(6);

        double tax=0.1;
        ServiceEstimate serviceEstimate = new ServiceEstimate(tax);
        serviceEstimate.setGroomingCost(100);
        serviceEstimate.setCareCharges(10);

        System.out.println(customer.toString());
        System.out.println(pet.toString());
        System.out.println("total money: ");
        System.out.println(serviceEstimate.getTotal());

    }
}
```

```java
public class Customer5 {
    String name;
    String adress;
    String phoneNumber;
    public String getName() {...}
    public void setName(String name) {...}
    public String getAddress() {...}
    public void setAdress(String adress) {...}
    public String getPhoneNumber() {...}
    public void setPhoneNumber(String phoneNumber) {...}
    @Override
    public String toString() {
        return "name: " + name + ", adress: " + adress + ", phoneNumber: " + phoneNumber;
    }

}
```

```java
public class Pet {
    int breek;
    int age;
    double weight;
    public int getBreek() {...}
    public void setBreek(int breek) {...}
    public int getAge() {...}
    public void setAge(int age) {...}
    public double getWeight() {...}
```

```java
public class ServiceEstimate {
    double groomingCost;
    double careCharges;
    double tax;
    public ServiceEstimate(double tax) {
        this.tax = tax;
    }
    public double getGroomingCost() {...}
    public void setGroomingCost(double groomingCost) {...}
```

```java
    public void setWeight(double weight) {...}
    @Override
    public String toString() {
        return "breek: " + breek + ", age: " + age + ", weight: " + weight;
    }

}
```

```java
    public double getCareCharges() {...}
    public void setCareCharges(double careCharges) {...}
    public double getTotal(){
        double total=(groomingCost+careCharges)*(1+tax);
        return total;
    }
}
```

Problem 6

```java
public class Problem6 {
    public static void main(String[] args) {
        Customer6 customer = new Customer6();
        customer.setName("a");
        customer.setAdrress("123/1 binh duong");
        customer.setPhoneNumber("0912345678");

        Records records = new Records();
        records.setTypeCake("birthday cake");
        records.setNumTiers(3);
        records.setEventDay("13/11/2019");

        double tax=0.1;
        PriceQuote priceQuote = new PriceQuote(tax);
        priceQuote.setCost(100);
        priceQuote.setLaborCharges(30);
        priceQuote.setDeliveryFee(10);

        System.out.println(customer.toString());
        System.out.println(records.toString());
        System.out.println("total money: ");
        System.out.println(priceQuote.getTotal());

    }
}
```

```java
public class Customer6 {
    String name;
    String adrress;
    String phoneNumber;
    public String getName() {...}
    public void setName(String name) {...}
    public String getAdrress() {...}
    public void setAdrress(String adrress) {...}
    public String getPhoneNumber() {...}
    public void setPhoneNumber(String phoneNumber) {...}
    @Override
    public String toString() {
        return "name: " + name + ", adrress: " + adrress + ", phoneNumber: " + phoneNumber;
    }

}
```

```java
public class Records {
    String typeCake;
    int numTiers;
    String eventDay;

    public String getTypeCake() {...}

    public void setTypeCake(String typeCake) {...}

    public int getNumTiers() {...}

    public void setNumTiers(int numTiers) {...}

    public String getEventDay() {...}

    public void setEventDay(String eventDay) {...}

    @Override
    public String toString() {
        return "typeCake: " + typeCake + ", numTiers: " + numTiers + ", eventDay: " + eventDay;
    }

}
```

```java
public class PriceQuote {
    double cost;
    double laborCharges;
    double deliveryFee;
    double tax;
    public PriceQuote(double tax) {
        this.tax = tax;
    }

    public double getCost() {...}

    public void setCost(double cost) {...}

    public double getLaborCharges() {...}

    public void setLaborCharges(double laborCharges) {...}

    public double getDeliveryFee() {...}

    public void setDeliveryFee(double deliveryFee) {...}
    public double getTotal(){
        double total = (cost+laborCharges+deliveryFee)*(1+tax);
        return total;
    }

}
```

Problem 7

```java
public class Problem7 {
    public static void main(String[] args) {
        Customer7 customer = new Customer7();
        customer.setName("a");
        customer.setAdrress("123/1 binh duong");
        customer.setLicenseNumber("0912345678");

        Cars cars = new Cars();
        cars.setMake("BMW");
        cars.setModel("i8");
        cars.setRentalPeriod(5);
        cars.setMileageLimit(4000);

        double tax = 0.1;
        RentalAgreement rentalAgreement = new RentalAgreement(tax);
        rentalAgreement.setRentalFee(1000);
        rentalAgreement.setMileageFees(100);

        System.out.println(customer.toString());
        System.out.println(cars.toString());
        System.out.println("Total Rental : ");
        System.out.println(rentalAgreement.getTotalRentalCost());

    }
}
```

```java
public class Customer7 {
    String name;
    String adrress;
    String licenseNumber;
    public String getName() {...}
    public void setName(String name) {...}
    public String getAdrress() {...}
    public void setAdrress(String adrress) {...}
    public String getLicenseNumber() {...}
    public void setLicenseNumber(String phoneNumber) {...}
    @Override
    public String toString() {
        return "name: " + name + ", adrress: " + adrress + ", license number: " + licenseNumber;
    }

}
```

```java
public class Cars {
    String make;
    String model;
    int rentalPeriod;
    int mileageLimit;

    public String getMake() {...}
    public void setMake(String make) {...}
    public String getModel() {...}
    public void setModel(String model) {...}
    public int getRentalPeriod() {...}
    public void setRentalPeriod(int rentalPeriod) {...}
    public int getMileageLimit() {...}
    public void setMileageLimit(int mileageLimit) {...}

    @Override
    public String toString() {
        StringBuilder sb = new StringBuilder();
        sb.append("make: ").append(make);
        sb.append(", model: ").append(model);
        sb.append(", rentalPeriod: ").append(rentalPeriod);
        sb.append(", mileageLimit: ").append(mileageLimit);
        return sb.toString();
    }

}
```

```java
public class RentalAgreement {
    double rentalFee;
    double mileageFees;
    double tax;

    public RentalAgreement(double tax) {
        this.tax = tax;
    }
    public double getRentalFee() {...}

    public void setRentalFee(double rentalFee) {...}

    public double getMileageFees() {...}

    public void setMileageFees(double mileageFees) {...}
    public double getTotalRentalCost(){
        double total = (rentalFee+mileageFees)*(1+tax);
        return total;
    }
}
```

Problem 8

```java
public class Problem8 {
    public static void main(String[] args) {
        Customer8 customer = new Customer8();
        customer.setName("a");
        customer.setAdrress("123/1 binh duong");
        customer.setPhoneNumber("0912345678");

        Session session = new Session();
        session.setType("wedding");
        session.setDuration(12);
        session.setDate("13/11/2019");

        double tax=0.1;
        BookingConfirmation booking = new BookingConfirmation(tax);
        booking.setSessionFee(100);
        booking.setAdditionalCharges(10);

        System.out.println(customer.toString());
        System.out.println(session.toString());
        System.out.println("total price: ");
        System.out.println(booking.getTotalPrice());

    }
}
```

```java
public class Customer8 {
    String name;
    String adrress;
    String phoneNumber;
    public String getName() {...}
    public void setName(String name) {...}
    public String getAdrress() {...}
    public void setAdrress(String adrress) {...}
    public String getPhoneNumber() {...}
    public void setPhoneNumber(String phoneNumber) {...}
    @Override
    public String toString() {
        return "name: " + name + ", adrress: " + adrress + ", phoneNumber: " + phoneNumber;
    }

}
```

```java
public class Session {
    String type;
    int duration;
    String date;

    public String getType() {...}

    public void setType(String type) {...}
```

```java
public class BookingConfirmation {
    double sessionFee;
    double additionalCharges;
    double tax;

    public BookingConfirmation(double tax) {
        this.tax = tax;
    }
```

```java
    public int getDuration() {}

    public void setDuration(int duration) {}

    public String getDate() {}

    public void setDate(String date) {}

    @Override
    public String toString() {
        StringBuilder sb = new StringBuilder();
        sb.append("type: ").append(type);
        sb.append(", duration: ").append(duration);
        sb.append(", date: ").append(date);
        return sb.toString();
    }

}
```

```java
    public double getSessionFee() {}

    public void setSessionFee(double sessionFee) {}

    public double getAdditionalCharges() {}

    public void setAdditionalCharges(double additionalCharges) {}

    public double getTotalPrice(){
        double total = (sessionFee+additionalCharges)*(1+tax);
        return total;
    }
}
```

Problem 9

```java
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    ListUser listUser = new ListUser();
    ListBook listBook = new ListBook();
    User user1 = new User();
    user1.setId("1234");
    user1.setName("a");
    user1.setEmail("a@gmail.com");
    Book book = new Book();
    System.out.println("Enter ISBN: ");
    book.setISBN(sc.nextLine());
    System.out.println("Enterre litle: ");
    book.setTitle(sc.nextLine());
    System.out.println("Enterre number of days they want to borrow: ");
    int numDay = sc.nextInt();
    if(!listBook.Check(book)){
        System.out.println("Library dont have this book");
    }
    else if(!listUser.Check(user1)){
        if(listUser.getOverdueBook()>1){
            System.out.println("user is holding "+listUser.getOverdueBook()+" overdue book");
        }
        else{
            System.out.println("Book is available but wrong user");
        }
    }
    else{
        System.out.println("you can borrow it for "+numDay+" days");
```

```java
public class Book {
    String ISBN;
    String title;
    public String getISBN() {
        return ISBN;
    }
    public void setISBN(String iSBN) {
        ISBN = iSBN;
    }
    public String getTitle() {
        return title;
    }
    public void setTitle(String title) {
        this.title = title;
    }
    public String getValue(){
        return ISBN + " " + title;
    }

}
```

```java
public class ListBook {
    Book[] arrBook = new Book[3];

    public ListBook(){
        this.arrBook[0] = new Book();
        this.arrBook[0].setISBN("a12");
        this.arrBook[0].setTitle("toan");
        this.arrBook[1] = new Book();
        this.arrBook[1].setISBN("a13");
        this.arrBook[1].setTitle("van");
        this.arrBook[2] = new Book();
        this.arrBook[2].setISBN("a14");
        this.arrBook[2].setTitle("anh");

    }

    public boolean Check(Book book){
        String a = book.getValue();
        for(int i=0; i<arrBook.length; i++){
            String b = arrBook[i].getValue();
            if(a.equals(b)){
                    return true;
            }
        }
        return false;
    }
}
```

```java
public class User {
    String id;
    String name;
    String email;

    public String getId() {}

    public void setId(String id) {}

    public String getName() {}

    public void setName(String name) {}

    public String getEmail() {}

    public void setEmail(String email) {}

    public String getValue(){
        return id + " " + name + " " + email;
    }

}
```

```java
public class ListUser {
    User[] arrUser = new User[3];
    int[] overdueBook = new int[3];
    int countOverdue;
    public ListUser(){
        this.arrUser[0] = new User();
        this.arrUser[0].setId("1234");
        this.arrUser[0].setName("a");
        this.arrUser[0].setEmail("a@gmail.com");
        this.overdueBook[0]=0;
        this.arrUser[1] = new User();
        this.arrUser[1].setId("1235");
        this.arrUser[1].setName("b");
        this.arrUser[1].setEmail("b@gmail.com");
        this.overdueBook[1]=1;
        this.arrUser[2] = new User();
        this.arrUser[2].setId("1236");
        this.arrUser[2].setName("c");
        this.arrUser[2].setEmail("c@gmail.com");
        this.overdueBook[2]=2;
    }
    public int getOverdueBook(){
        return countOverdue;
    }
    public boolean Check(User user){
        String a = user.getValue();
        for(int i=0; i<arrUser.length; i++){
            String b = arrUser[i].getValue();
```

```java
        this.arrUser[2] = new User();
        this.arrUser[2].setId("1236");
        this.arrUser[2].setName("c");
        this.arrUser[2].setEmail("c@gmail.com");
        this.overdueBook[2]=2;
    }
    public int getOverdueBook(){
        return countOverdue;
    }
    public boolean Check(User user){
        String a = user.getValue();
        for(int i=0; i<arrUser.length; i++){
            String b = arrUser[i].getValue();
            if(a.equals(b)){
                if(overdueBook[i]<=1)
                    return true;
                else{
                    this.countOverdue+=overdueBook[i];
                }
            }
        }
        return false;
    }

}
```